# Diamond Price Prediction: Midterm Report

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## 1 Summary of Findings

The project models diamond price as a supervised regression problem using the standard diamonds dataset. At the midterm stage, the strongest model is XGBoost: it reaches a 5-fold cross-validated RMSE of USD 526.9 ± 10.3 and MAE of USD 262.1 on the training split. Random Forest is close, with RMSE USD 552.5 ± 9.6, while Linear Regression is much weaker at RMSE USD 1,467.4 ± 153.5. All trained models substantially improve on the mean-price baseline.

The main modeling finding is that nonlinear tree ensembles fit this dataset far better than a linear model. The gain from Random Forest to XGBoost is smaller than the gain from Linear Regression to Random Forest, so XGBoost is the current leader but Random Forest remains a credible alternative. The held-out test set has not been evaluated in this midterm version; it is reserved for the final evaluation after model selection.

## 2 Dataset Findings

The raw dataset contains 53,940 rows and 10 columns: `price`, `carat`, `cut`, `color`, `clarity`, `depth`, `table`, `x`, `y`, and `z`. The target is strictly positive, ranging from USD 326 to USD 18,823. No missing values were found in the required columns.

Cleaning removed 168 rows, or approximately 0.31% of the raw dataset. The removed records consisted of 20 rows with zero physical dimensions, 3 extreme dimension outliers, and 145 duplicate rows after invalid dimensional records were excluded. This is a small removal rate, so the cleaning step improves data validity without materially reducing the size of the training sample. The final cleaned dataset contains 53,772 rows.

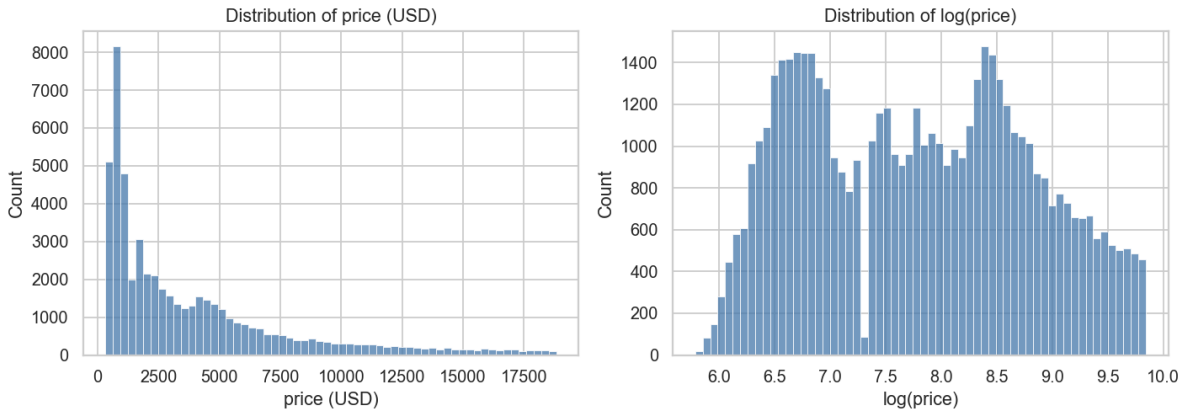



Figure 1: Distribution of price and log(price).

The target distribution is strongly right-skewed. Most diamonds are comparatively low-priced, while a smaller number of high-value diamonds extend the upper tail. This supports the use of a log-transformed target during model fitting: it reduces the dominance of expensive diamonds during training while still allowing final errors to be reported in USD after inverse transformation.

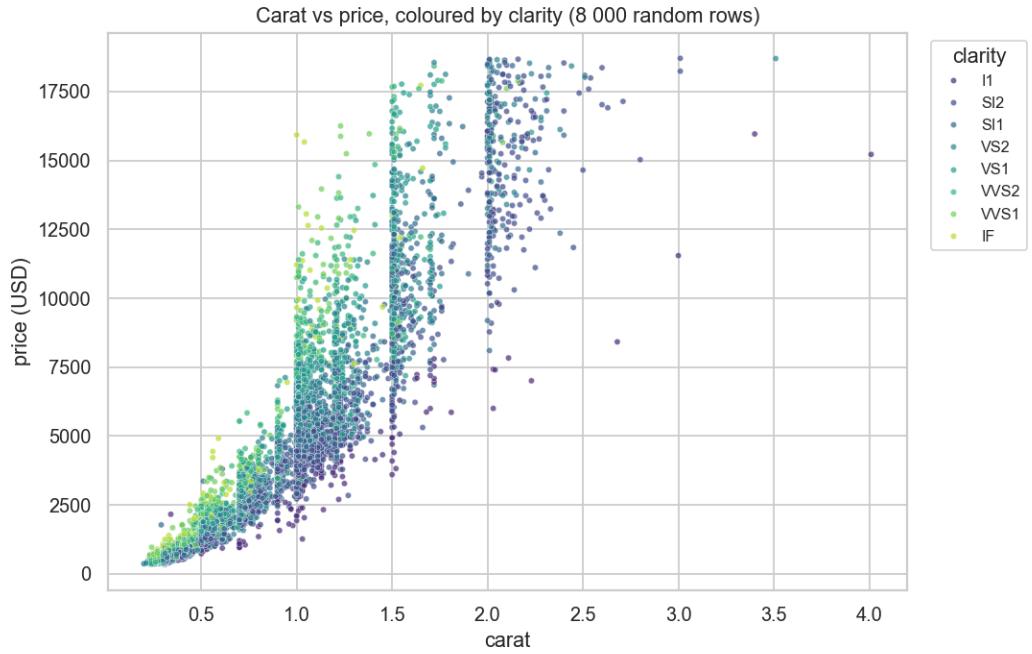



Figure 2: Carat versus price, sampled for readability and coloured by clarity.

Carat has the clearest visible relationship with price. The relationship is nonlinear: prices rise slowly for small stones, then increase more sharply for larger stones. The spread also increases as carat increases, indicating that quality grades and proportions matter more at higher price levels. This pattern is consistent with the later model results: a purely linear model captures the general trend, but tree ensembles capture nonlinear interactions more effectively.

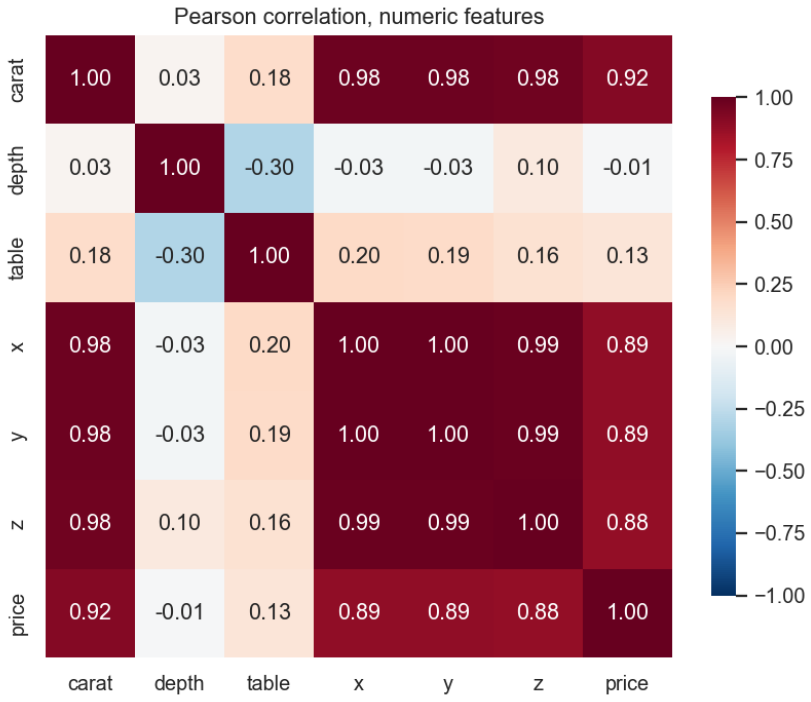



Figure 3: Correlation heatmap for numeric features.

The correlation matrix shows that `carat`, `x`, `y`, and `z` are highly collinear. They all describe stone size from different angles. This is useful for prediction but complicates interpretation, because a model can assign importance to one size proxy even when the underlying signal is shared across several correlated variables.

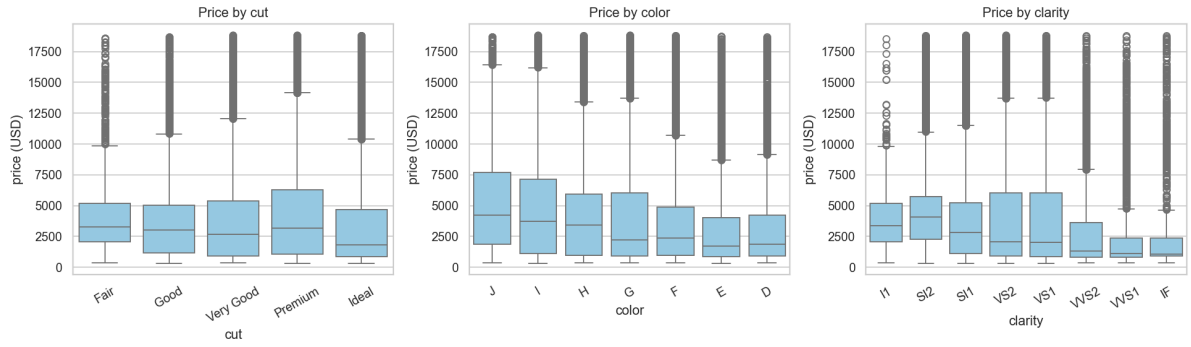



Figure 4: Price distributions by cut, color, and clarity.

The categorical quality grades show visible differences in price distributions, but their raw boxplots are affected by the distribution of stone size inside each category. For example, a high-quality category does not automatically have the highest raw median price if it contains many smaller stones. These plots therefore support including the categorical variables, but they do not by themselves establish a causal ranking of price effects.

## 3 Modeling Approach

The cleaned dataset was split into 80% training data and 20% held-out test data using a fixed random seed and stratification on carat quintiles. All midterm model comparison uses only the training split through 5-fold cross-validation. The test split remains unused at this stage.

The preprocessing pipeline encodes `cut`, `color`, and `clarity` as ordinal variables using their domain order. The `carat` feature is transformed with `log1p`. Numeric features are standardized for Linear Regression and passed through unscaled for tree-based models. The target is log-transformed during fitting and inverse-transformed before metric computation, so all reported errors are on the original USD scale.

Four models are compared: a mean-price baseline, Linear Regression, Random Forest with 200 trees, and XGBoost with 400 trees. Hyperparameter tuning is not part of the midterm result; the purpose here is to establish a reproducible baseline and compare the default model families fairly under the same cross-validation protocol.

## 4 Model Results

Table 1: Training-split 5-fold cross-validation results. Lower RMSE, MAE, and MAPE are better; higher $R^2$ is better.

| Model | RMSE (USD) | MAE (USD) | $R^2$ | MAPE |
|---|---|---|---|---|
| Mean baseline | 3971.6 ± 42.2 | 3019.9 | -0.0002 | 187.5% |
| Linear Regression | 1467.4 ± 153.5 | 559.5 | 0.8618 | 12.7% |
| Random Forest | 552.5 ± 9.6 | 271.8 | 0.9806 | 6.5% |
| XGBoost | **526.9 ± 10.3** | **262.1** | **0.9824** | **6.1%** |

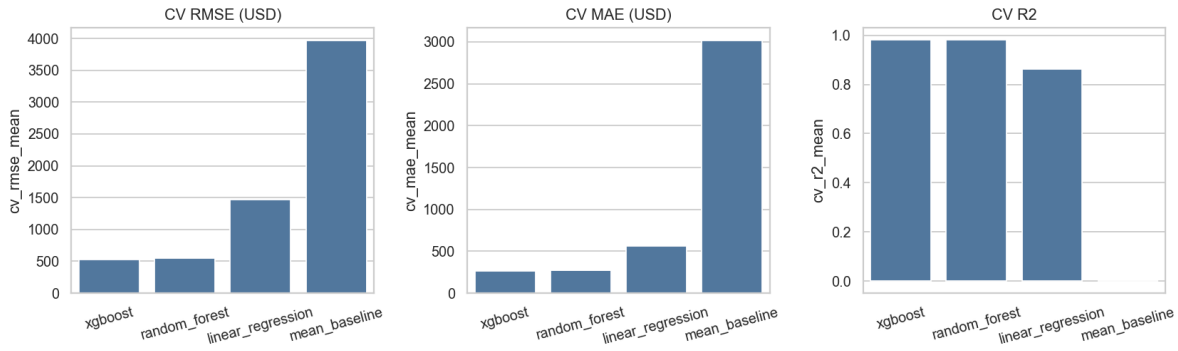



Figure 5: Cross-validation metric comparison.

The mean baseline performs poorly, with RMSE close to USD 4,000 and negative $R^2$, confirming that predicting the same average price for every diamond is not informative. Linear Regression reduces RMSE by about 63% compared with the baseline, which shows that the encoded features carry strong predictive signal. However, its remaining error is still high, especially compared with the tree ensembles.

Random Forest reduces RMSE by a further 62% compared with Linear Regression. This is the main performance jump in the current experiment and indicates that nonlinear splits and feature interactions are central to the task. XGBoost gives the best overall result, reducing RMSE by another 4.6% compared with Random Forest and achieving the best MAE, $R^2$, and

MAPE. The improvement over Random Forest is real in the current cross-validation table, but it is modest. The two tree models remain close enough that both are relevant for final tuning.

## 5 Learning Behavior and Error Analysis

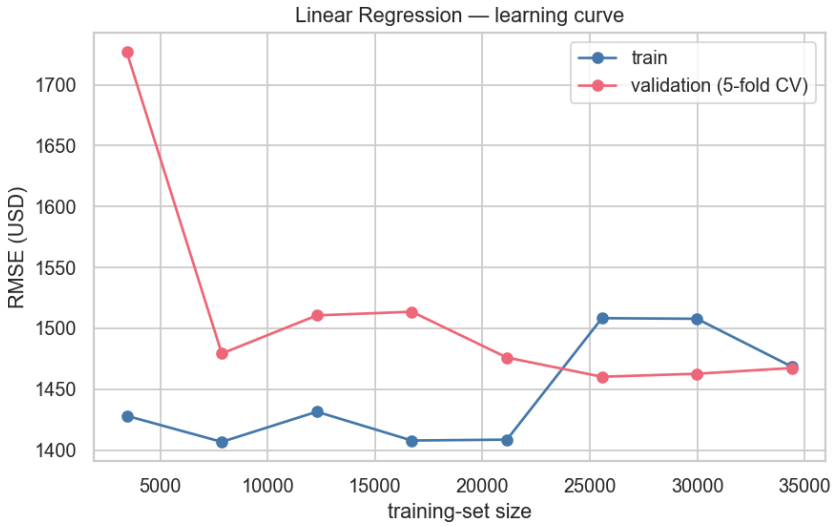



Figure 6: Linear Regression learning curve.

The Linear Regression learning curve stabilizes at a validation RMSE near USD 1,467. Adding more training data does not close the gap to the tree ensembles, which points to limited model capacity rather than insufficient sample size. This matches the EDA: the price relationship is nonlinear and depends on interactions between size, proportions, and quality grades.

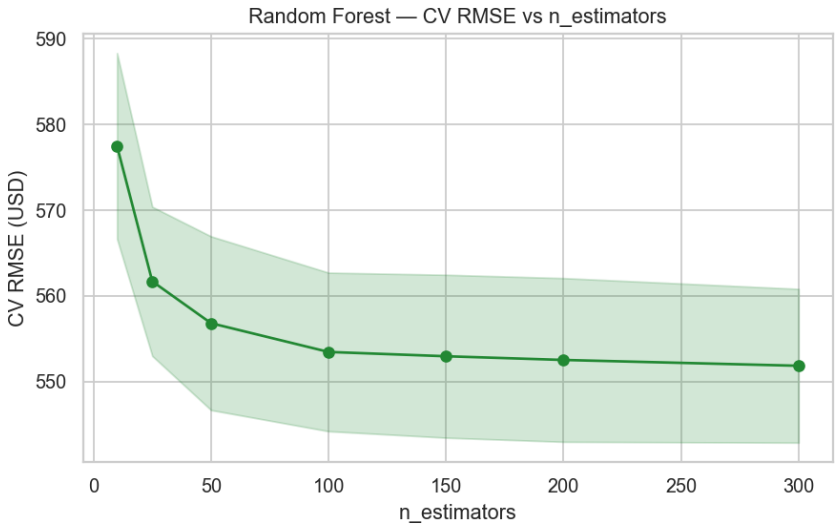



Figure 7: Random Forest cross-validated RMSE versus number of trees.

Random Forest improves quickly as the number of trees increases, then the curve flattens around the 100–200 tree range. The midterm setting of 200 trees is therefore a reasonable default: it captures most of the ensemble benefit without unnecessary training cost.

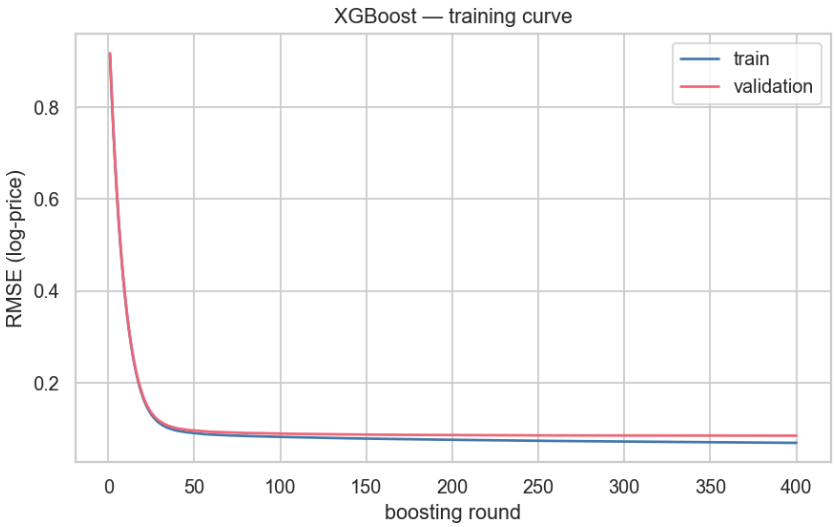



Figure 8: XGBoost training and validation RMSE over boosting rounds.

The XGBoost curve shows continuing reduction in training error and a weaker validation improvement after the earlier boosting rounds. The number of boosting rounds and the regularization settings are therefore important tuning parameters for the final phase. The current model is already strong, but the curve indicates room to make it more controlled.

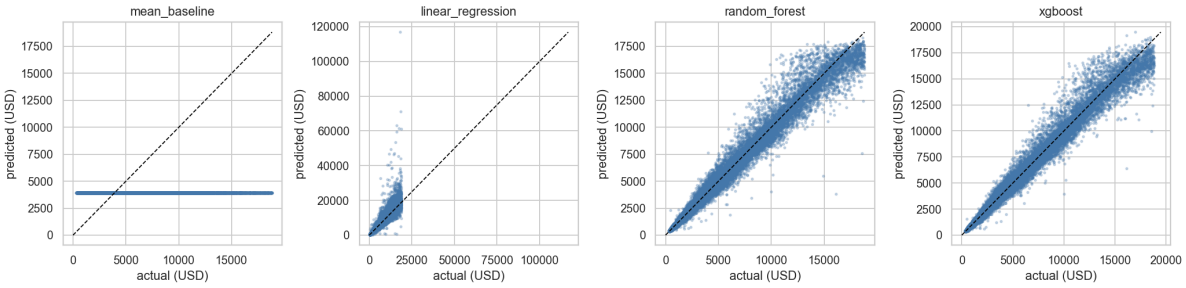



Figure 9: Cross-validated predicted versus actual prices.

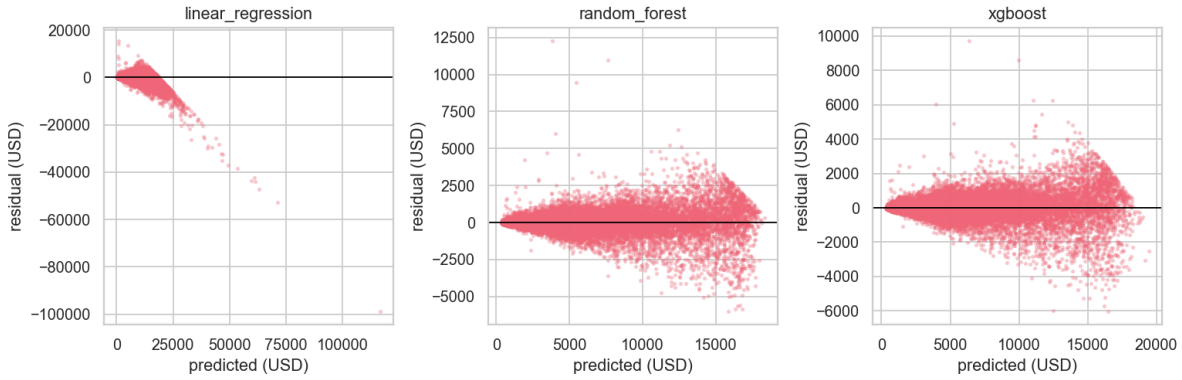



Figure 10: Cross-validated residuals for fitted models excluding the mean baseline.

The predicted-versus-actual plots confirm the ranking from the metric table. Tree ensemble predictions stay closer to the diagonal over most of the price range, while Linear Regression shows wider deviations. The residual plots also show larger absolute errors for more expensive diamonds. This is expected for a monetary target with a wide range: even after log-target fitting, high-value diamonds remain harder to predict precisely in dollar terms.

## 6 Feature Importance Discussion

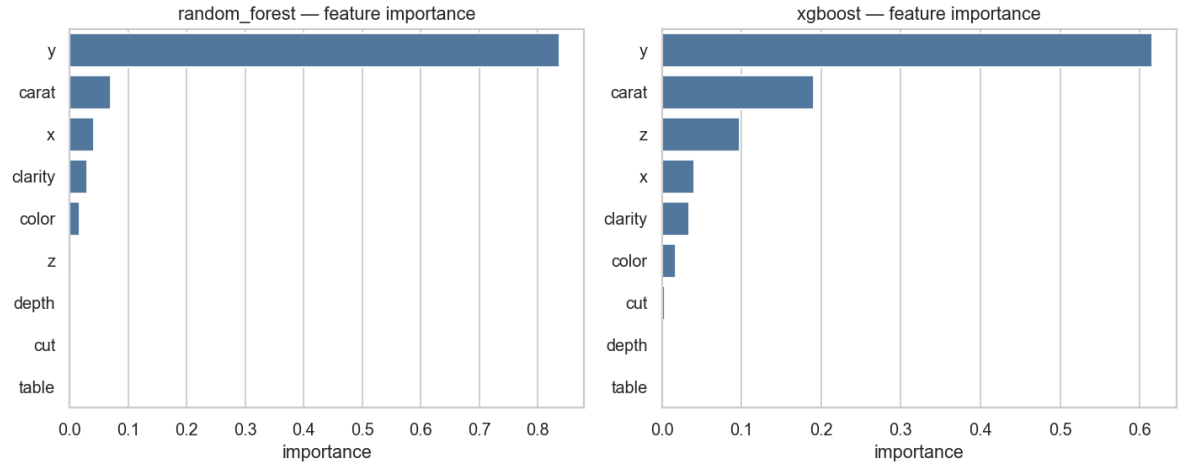



Figure 11: Model-native feature importance for Random Forest and XGBoost.

The model-native importance values are useful for diagnosis but not for causal interpretation. Random Forest assigns 83.7% of its importance to `y`, while XGBoost assigns 61.5% to `y`. `carat` is second, with 7.0% importance in Random Forest and 19.1% in XGBoost. This does not mean that diamond width is the main economic driver of price. It reflects the collinearity seen in the EDA: `carat`, `x`, `y`, and `z` all encode stone size, and tree models can concentrate shared size information into one proxy variable.

The practical interpretation is that size dominates price prediction, but the exact feature receiving the highest model-native importance is unstable because several size variables overlap. The strongest final interpretation will focus on grouped predictors such as size, proportions, and quality grades instead of overclaiming individual size-feature effects.

## 7 Limitations and Final-Phase Implications

The current results are midterm results, not final test results. The held-out test split has intentionally not been evaluated, preserving it as an unbiased final check after model selection and tuning. The current model comparison also uses fixed default hyperparameters, so the exact ranking may still change after controlled tuning on the training split.

Three issues are most important for the final phase. First, the best current model is only modestly better than Random Forest, so both tree ensembles deserve tuning before final selection. Second, XGBoost shows signs that the number of boosting rounds and regularization need closer control. Third, feature importance requires careful reporting because collinear size variables make single-feature explanations unreliable.

# 8 Conclusion

The midterm work establishes a reproducible prediction pipeline and produces a clear model ranking from cross-validation. The strongest result is XGBoost with RMSE USD 526.9 $\pm$ 10.3, MAE USD 262.1, $R^2 = 0.9824$, and MAPE 6.1%. The evidence supports using tree-based nonlinear models for this dataset, with XGBoost as the current selected checkpoint and Random Forest as a close alternative for the final tuning phase.